

# Correlation & Business Statistics

## Question 1

If students study more hours, will they score better?

Can we prove it using data?

Statistics helps answer this.

## Question 2

Suppose a company spends more money on advertising.

Will sales increase?

Maybe.

Statistics helps us measure this relationship.

This relationship is called **Correlation**.

# What is Correlation?

Correlation measures

How strongly two variables move together.

It tells us

- Direction
- Strength

It DOES NOT tell us

- Cause
- Effect

## Examples

| Variable X         | Variable Y      |
|--------------------|-----------------|
| Study Hours        | Exam Marks      |
| Temperature        | Ice Cream Sales |
| Advertising Budget | Sales           |
| Experience         | Salary          |

## Correlation Answers Three Questions

### 1. Are variables related?

Yes or No 

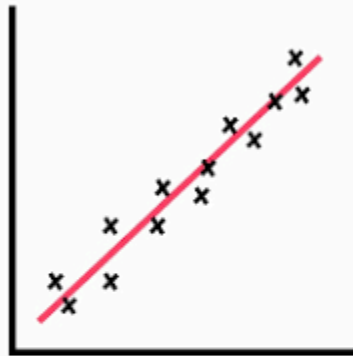
### 2. If related, in which direction?

Positive  
or  
Negative

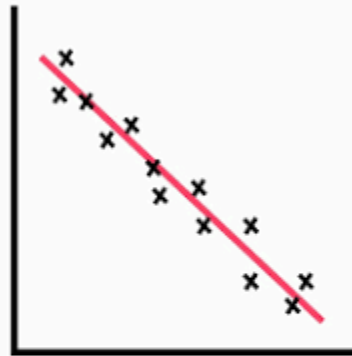
### 3. How strong is the relationship?

Weak  
Moderate  
Strong 

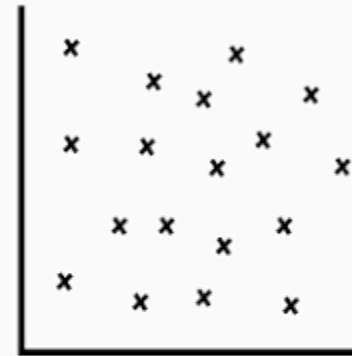
## Types of Correlation



Positive  
Correlation



Negative  
Correlation



No  
Correlation

### Positive Correlation

When one variable increases,  
the other also increases.

## Example

- More study - Higher marks
- More experience - Higher salary
- More advertisements - More sales

## Negative Correlation

One variable increases

Other decreases

### Example

- Price  $\uparrow$  - Demand  $\downarrow$
- Exercise  $\uparrow$  - Body Fat  $\downarrow$

## Zero Correlation

No relationship

### Example

- Shoe size
- Exam Marks

No meaningful relation.

# Correlation Coefficient

Represented by :  $r$

Range:  $-1 \leq r \leq +1$

## Interpretation Table

| r Value | Meaning              |
|---------|----------------------|
| +1      | Perfect Positive     |
| +0.9    | Very Strong Positive |
| +0.7    | Strong Positive      |
| +0.5    | Moderate             |
| +0.3    | Weak                 |
| 0       | No correlation       |
| -0.3    | Weak Negative        |
| -0.5    | Moderate Negative    |

|      |                      |
|------|----------------------|
| -0.7 | Strong Negative      |
| -0.9 | Very Strong Negative |
| -1   | Perfect Negative     |

## Pearson Correlation

The most commonly used correlation method.

Used to measure Linear relationship between two numerical variables.

### When to Use Pearson

Use when

- ✓ Continuous data
- ✓ Linear relationship
- ✓ Normally distributed data
- ✓ No significant outliers

## Pearson Formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

### Step-by-Step Example

Study Hours vs Marks

| Hours(X) | Marks(Y) |
|----------|----------|
| 2        | 50       |
| 4        | 60       |
| 6        | 70       |
| 8        | 80       |



|    |    |
|----|----|
| 10 | 90 |
|----|----|

## Step 1

Find  $X^2$

$Y^2$

$XY$

| X  | Y  | $X^2$ | $Y^2$ | $XY$ |
|----|----|-------|-------|------|
| 2  | 50 | 4     | 2500  | 100  |
| 4  | 60 | 16    | 3600  | 240  |
| 6  | 70 | 36    | 4900  | 420  |
| 8  | 80 | 64    | 6400  | 640  |
| 10 | 90 | 100   | 8100  | 900  |

Totals

$\Sigma X = 30$

$\Sigma Y = 350$

$$\Sigma X^2 = 220$$

$$\Sigma Y^2 = 25500$$

$$\Sigma XY = 2300$$

$$n = 5$$

$$r = 1$$



## Advantages of Pearson

- Simple
- Widely used
- Measures strength
- Useful in ML
- Useful in Finance
- Useful in Business

## Limitations of Pearson

- Cannot capture Curved relationships
- Sensitive to Outliers
- Needs Normal distribution
- Works only on numerical values

# Spearman Correlation

## Definition

Spearman Correlation measures

**The strength of the relationship between two ranked variables.**

Instead of actual values, it compares Ranks.

Represented by

**$\rho$  (pronounced "rho")**

## When to Use Spearman

- Ordinal Data
- Ranks
- Ratings
- Customer Reviews
- Competition Positions
- Survey Responses
- Data with Outliers
- Non-normal data

## Examples

- Movie Rankings
- Hotel Rankings
- Employee Ranking
- Customer Satisfaction
- College Ranking
- Olympic Ranking

## Spearman Formula

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$\rho$  = Spearman's rank correlation coefficient

$d_i$  = difference between the two ranks of each observation

$n$  = number of observations

Two Teachers ranked students.

| Student | Teacher 1 Rank | Teacher 2 Rank |
|---------|----------------|----------------|
| A       | 1              | 2              |
| B       | 2              | 1              |
| C       | 3              | 4              |
| D       | 4              | 3              |
| E       | 5              | 5              |

Create Table

| Student | Rank1 | Rank2 | d  | d <sup>2</sup> |
|---------|-------|-------|----|----------------|
| A       | 1     | 2     | -1 | 1              |
| B       | 2     | 1     | 1  | 1              |
| C       | 3     | 4     | -1 | 1              |
| D       | 4     | 3     | 1  | 1              |
| E       | 5     | 5     | 0  | 0              |

$$\Sigma d^2 = 4$$

$$n = 5$$

Substitute

$$\rho = 1 - \frac{6(4)}{5(25-1)}$$

$$= 1 - \frac{24}{120}$$

$$= 1 - 0.2$$

$$= 0.8$$

Interpretation

Strong Positive Rank Correlation



## Why Spearman Uses Ranks?

Suppose salaries are

₹20,000

₹25,000

₹1,50,000

₹22,000

₹23,000

Pearson gets heavily influenced by ₹1,50,000 (an outlier).

Spearman first converts them into ranks.

Rank 1

Rank 2

Rank 5

Rank 3

Rank 4

Now the outlier has much less influence.

## Pearson vs Spearman

| Feature                      | Pearson    | Spearman            |
|------------------------------|------------|---------------------|
| Data                         | Continuous | Ordinal/Rank        |
| Relationship                 | Linear     | Monotonic           |
| Uses Actual Values           | Yes        | No                  |
| Uses Ranks                   | No         | Yes                 |
| Outlier Sensitive            | Yes        | No (less sensitive) |
| Normal Distribution Required | Yes        | No                  |
| Coefficient                  | r          | $\rho$ (rho)        |



## How to Choose?

None

Is the data numerical?



├─ No → Spearman



Is the relationship linear?



├─ No → Spearman



Is the data approximately normal and free of major outliers?



├─ No → Spearman



Use Pearson

## Business Applications

| Pearson                                                                                                                                                                                                              | Spearman                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Advertising vs Sales</li> <li>• Salary vs Experience</li> <li>• Income vs Spending</li> <li>• Temperature vs Electricity Usage</li> <li>• Stock Price vs Revenue</li> </ul> | <ul style="list-style-type: none"> <li>• Customer Ratings</li> <li>• Employee Rankings</li> <li>• Hotel Rankings</li> <li>• Survey Responses (Likert Scale)</li> <li>• Sports Rankings</li> </ul> |

## Machine Learning Applications

| Pearson                                                                                                                                                                  | Spearman                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Feature selection</li> <li>• Detect multicollinearity</li> <li>• Correlation heatmaps</li> <li>• Regression analysis</li> </ul> | <ul style="list-style-type: none"> <li>• Non-linear but monotonic relationships</li> <li>• Robust feature selection with outliers</li> <li>• Ranking-based recommendation systems</li> </ul> |

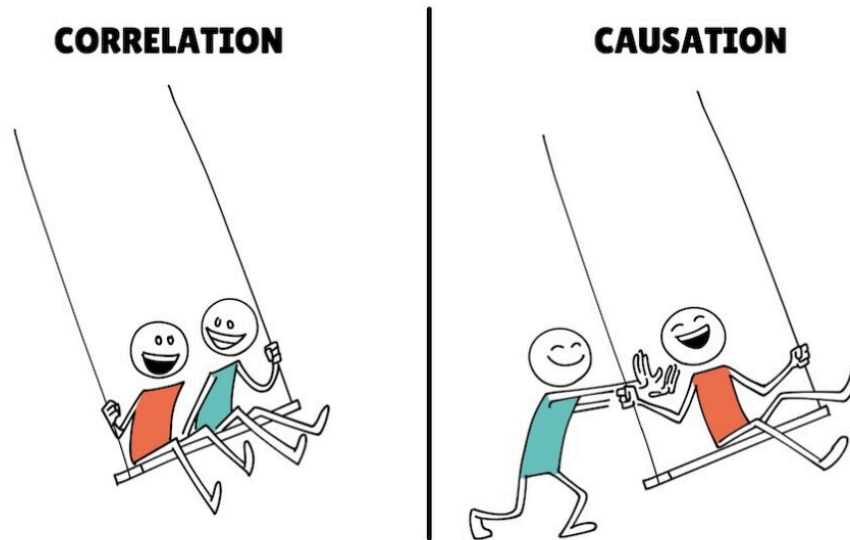
## Identify the Correct Correlation

| Scenario                          | Pearson or Spearman? |
|-----------------------------------|----------------------|
| Height vs Weight                  | ?                    |
| Customer Satisfaction (1–5 stars) | ?                    |
| Employee Rankings                 | ?                    |
| Advertising Spend vs Sales        | ?                    |
| Movie Ratings                     | ?                    |

## Summary

- **Pearson Correlation** measures the **strength and direction of a linear relationship** between two **continuous numerical variables**.
- **Spearman Correlation** measures the **strength and direction of a monotonic relationship** using **ranked data**.
- Use **Pearson** for continuous, approximately normally distributed data with a linear relationship.
- Use **Spearman** for ordinal/ranked data, non-normal data, or when outliers are present.
- Both coefficients range from **-1 to +1**, where the sign indicates direction and the magnitude indicates strength.
- Correlation measures **association**, but it does **not prove causation**.

# Correlation vs Causation



Suppose a company spends more money on advertising.

After one month, sales also increase.

**Did advertising cause the increase in sales?**

**What if a festive season (e.g., Diwali or Christmas) also occurred during that month?**

Could sales have increased because of the festival instead?

This introduces the idea that **correlation alone cannot prove cause and effect.**

# What is Causation?

Causation means:

**One variable directly causes a change in another variable.**

There is a **cause-and-effect relationship**.

Example

Studying more causes better exam performance.

Medicine lowers blood pressure.

Exercise reduces body fat.

Smoking increases the risk of lung cancer.

## Simple Definition

Correlation

Variables move together.

Causation

One variable directly produces a change in another.

# Difference Between Correlation and Causation

| Correlation                            | Causation                                                     |
|----------------------------------------|---------------------------------------------------------------|
| Shows association                      | Shows cause and effect                                        |
| Variables move together                | One variable changes another                                  |
| Does not prove cause                   | Proves cause                                                  |
| Easy to calculate                      | Requires experiments or strong evidence                       |
| Measured using correlation coefficient | Usually established through controlled studies or experiments |

## Ice Cream Sales vs Drowning Cases

| Summer Temperature | Ice Cream Sales | Drowning Cases |
|--------------------|-----------------|----------------|
| Low                | Low             | Low            |
| Medium             | Medium          | Medium         |
| High               | High            | High           |

Question

Does eating ice cream cause drowning?

Answer

No.

Both increase because of

**Summer**

Summer is called a

**Confounding Variable.**

## What is a Confounding Variable?

A **confounding variable** is an external factor that influences both variables, making it appear as though one causes the other.

## Umbrella Sales vs Rain

| Rainfall | Umbrella Sales |
|----------|----------------|
| Low      | Low            |
| Medium   | Medium         |
| High     | High           |

Question

Do umbrellas cause rain?

No.

Rain causes people to buy umbrellas.

## Hospital Visits vs Death Rate

More hospital visits : More deaths

Question

Do hospitals cause deaths?

Seriously ill people go to hospitals.

Illness is the actual cause.



## Shoe Size vs Reading Ability

Children with larger shoe sizes often read better.

Question

Does having bigger feet improve reading?

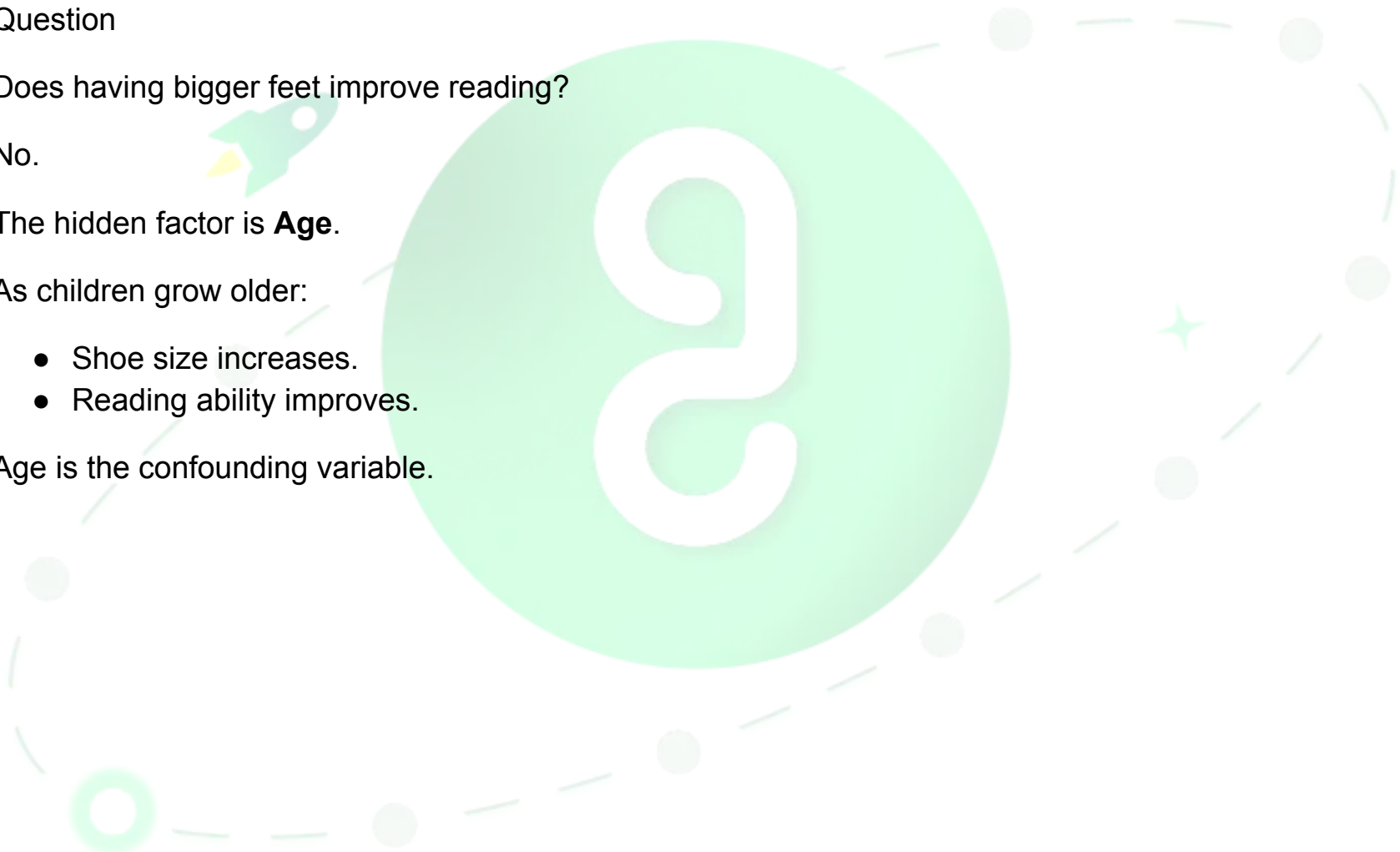
No.

The hidden factor is **Age**.

As children grow older:

- Shoe size increases.
- Reading ability improves.

Age is the confounding variable.



## Business Case Study

Suppose Amazon notices:

Customers who purchase coffee machines also purchase coffee mugs.

Question

Does buying a coffee machine cause customers to buy mugs?

Not necessarily.

The customer may simply be setting up a new kitchen.

The hidden factor is

**Customer Need.**

## Healthcare Example

Patients who exercise regularly usually have lower blood pressure.

Can exercise cause lower blood pressure?

Medical research strongly supports this.

Since controlled clinical studies have been conducted, this is considered a **causal relationship**.

# Machine Learning Perspective

Correlation is used for:

- Feature Selection
- Exploratory Data Analysis (EDA)
- Detecting Multicollinearity
- Finding relationships between variables

Machine learning models usually identify **patterns**, not **causes**.

Example

A model predicts that people buying baby products often buy diapers.

The model does not know **why**.

It only detects the association.

# Key Takeaways

- **Correlation** measures how strongly two variables move together.
- **Causation** means one variable directly produces a change in another.
- A strong correlation **does not prove** that one variable causes the other.
- Hidden factors (confounding variables), reverse causation, and coincidence can create misleading correlations.
- In business analytics and data science, **correlation is useful for discovering patterns**, while **causation requires stronger evidence**, such as A/B testing, controlled experiments, or long-term studies.
- Always ask: **"Is this relationship causal, or is there another explanation?"** This is a critical thinking skill for every data analyst and data scientist.